MCQ Generator

Q1. What is the initial acceleration of the CM of the stick in part (a) of the problem?

A. g
B. 2g/3
C. 3g/4
D. g/4

Q2. What is the result of the friction force on the cylinder in the given scenario?

A. The friction force produces a force component Tsinθ in the y direction.
B. The friction force produces a force component Tsinθ in the x direction.
C. The friction force produces a force component Tdθcosθ in the y direction.
D. The friction force produces a force component Tdθcosθ in the x direction.

Q3. If you slowly lengthen the string so that r doubles, then h increases by a factor of

A. 2
B. 4
C. 8
D. 16

Q4. If β = 0, then the results in both parts (a) and (b) reduce to

A. g/7
B. g/5
C. g/3
D. g/2

Q5. What is the point at which the chimney is most likely to break?

A. At the bottom of the chimney
B. At the top of the chimney
C. One-third of the way up the chimney
D. Two-thirds of the way up the chimney

Q6. What is the correct statement regarding the final relative velocity in the ball and stick theorem?

A. The final relative velocity is the same as the initial relative velocity.
B. The final relative velocity is the negative of the initial relative velocity.
C. The final relative velocity is greater than the initial relative velocity.
D. The final relative velocity is less than the initial relative velocity.

Q7. What is the minimum initial speed required for a wheel with all the mass on its rim to climb up over the step?

A. √(2gH)
B. √(2gH) / 2
C. √(2gH) / (5/2)
D. √(2gH) / (7/2)

Q8. What is the relation between the friction force and the angular acceleration?

A. $F = -\beta R\alpha$
B. $F = \beta R\alpha$
C. $F = \alpha R\beta$
D. $F = -\alpha R\beta$

Q9. What is the ratio of the speeds of the bottom ends of the sticks?

A. 1
B. $2(1-\beta)/(1+\beta)$
C. β
D. $1/\beta$

Q10. What is the main idea of Chasles' theorem?

A. It describes the general form of any motion in a rigid body.
B. It states that a rigid body can be transformed into itself by any manner.
C. It claims that any transformation of a rigid sphere has the property that there exist two points that end up where they started.
D. It explains why rotations can be hard to visualize.